

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers worldwide

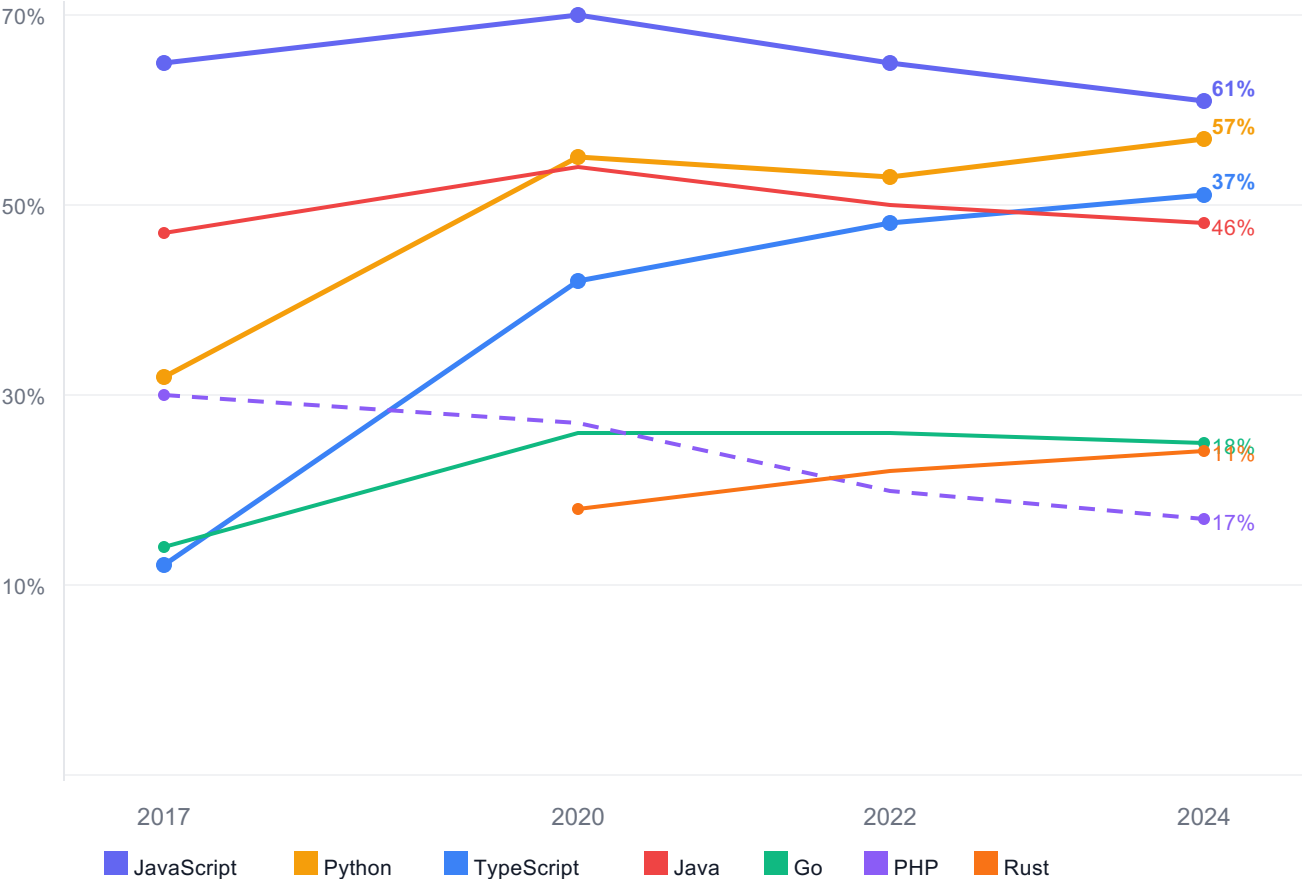
JETBRAINS DEVELOPER ECOSYSTEM REPORT

23,262

Developers Surveyed
Worldwide · 2024

The top-5 language landscape is reshuffling — Python nearly matches JavaScript, TypeScript triples, PHP halves

Python Closes the Gap: 8 Years of Language Usage Shifts



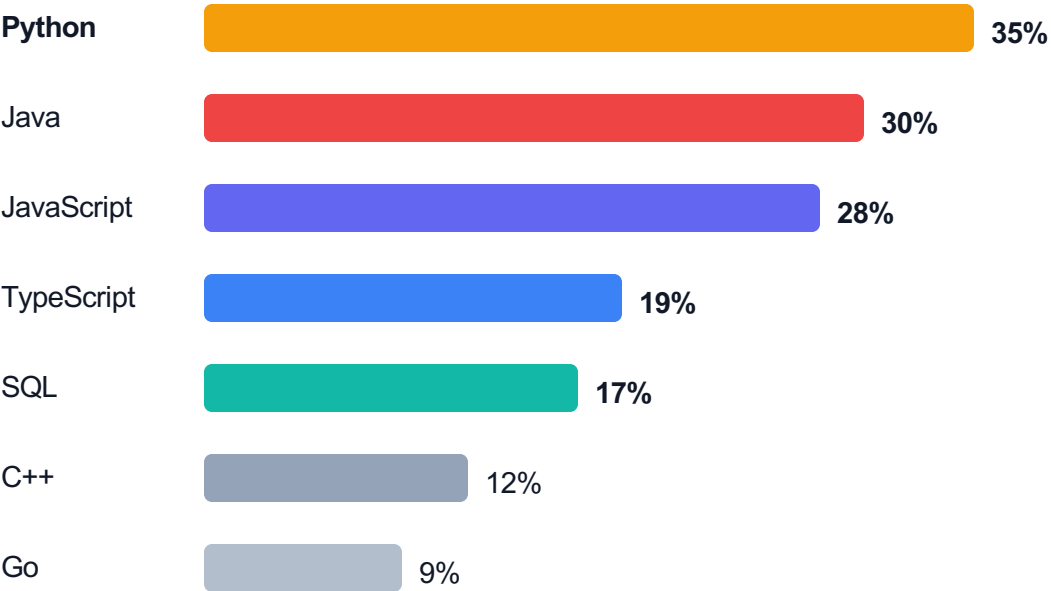
Language	2017	2020	2022	2024	Trend
JavaScript	65%	70%	65%	61%	▲ -4pp
Python	32%	55%	53%	57%	▲ +25pp
HTML/CSS	60%	61%	54%	51%	▲ -9pp
SQL	42%	56%	49%	48%	Flat
Java	47%	54%	48%	46%	▲ -1pp
TypeScript	12%	28%	34%	37%	▲ +25pp
Go	8%	19%	19%	18%	Flat
Rust	—	7%	9%	11%	▲ +4pp
PHP	30%	27%	20%	17%	▲ -13pp
Kotlin	2%	17%	16%	14%	Flat

Key shift: Python rose +25pp since 2017 while JavaScript fell from its 2020 peak. TypeScript is the fastest-growing typed language.

Python leads primary language adoption at 35%, but switching intent stays below 6% — ecosystem lock-in is real

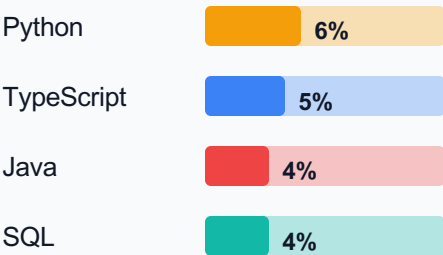
Python Is Now the #1 Primary Language at 35%

PRIMARY LANGUAGE SHARE



Adoption Intent: Planning to Switch

% of developers planning to adopt as primary language



INSIGHT

Developers rarely switch their primary stack.
Adoption intent is below 6% for every language —
expertise investment and ecosystem lock-in dominate.

<6%

Max adoption intent
across all languages

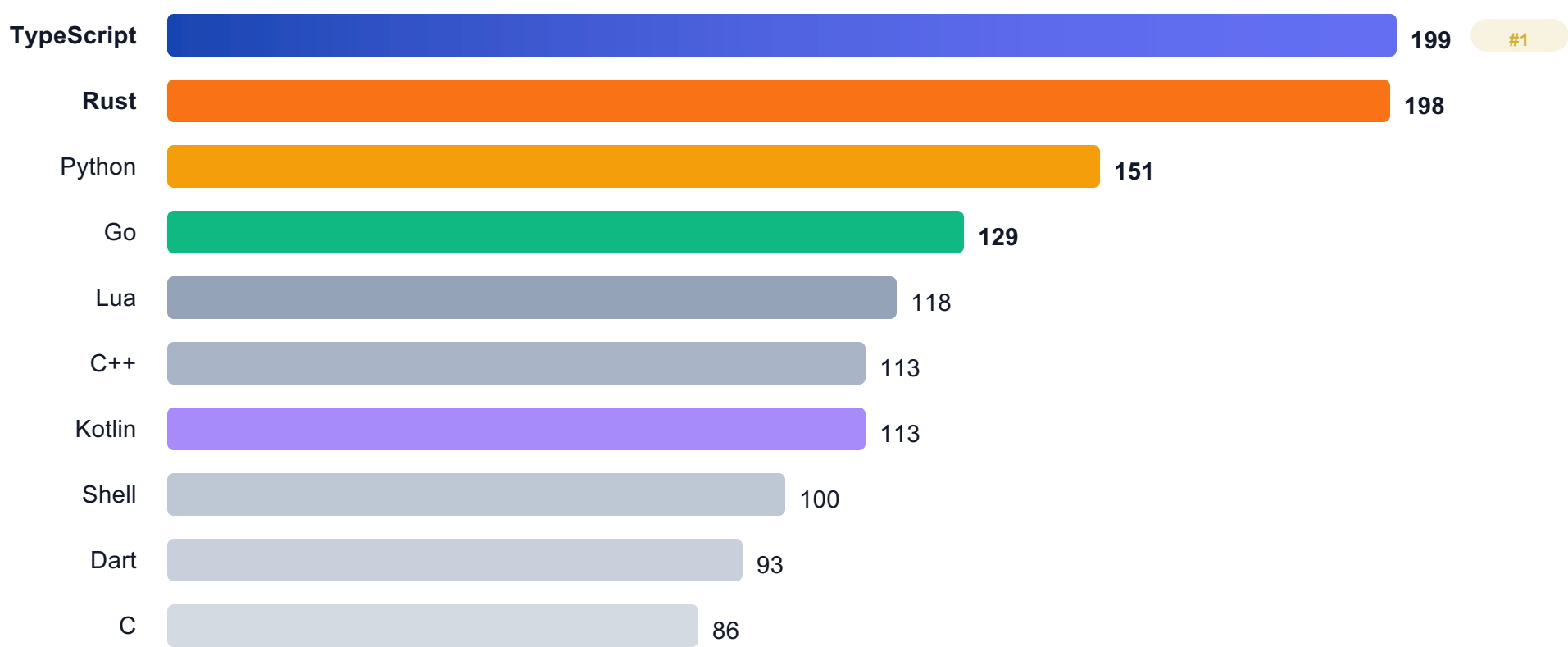
35%

Python as primary
#1 most chosen

Type-safe languages with modern tooling dominate satisfaction — TypeScript (199) and Rust (198) lead

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Developer satisfaction score (higher = more loved)



What They Share

- ✓ Type safety
- ✓ Modern tooling
- ✓ Expressive types
- ✓ Strong ecosystems

The Gap

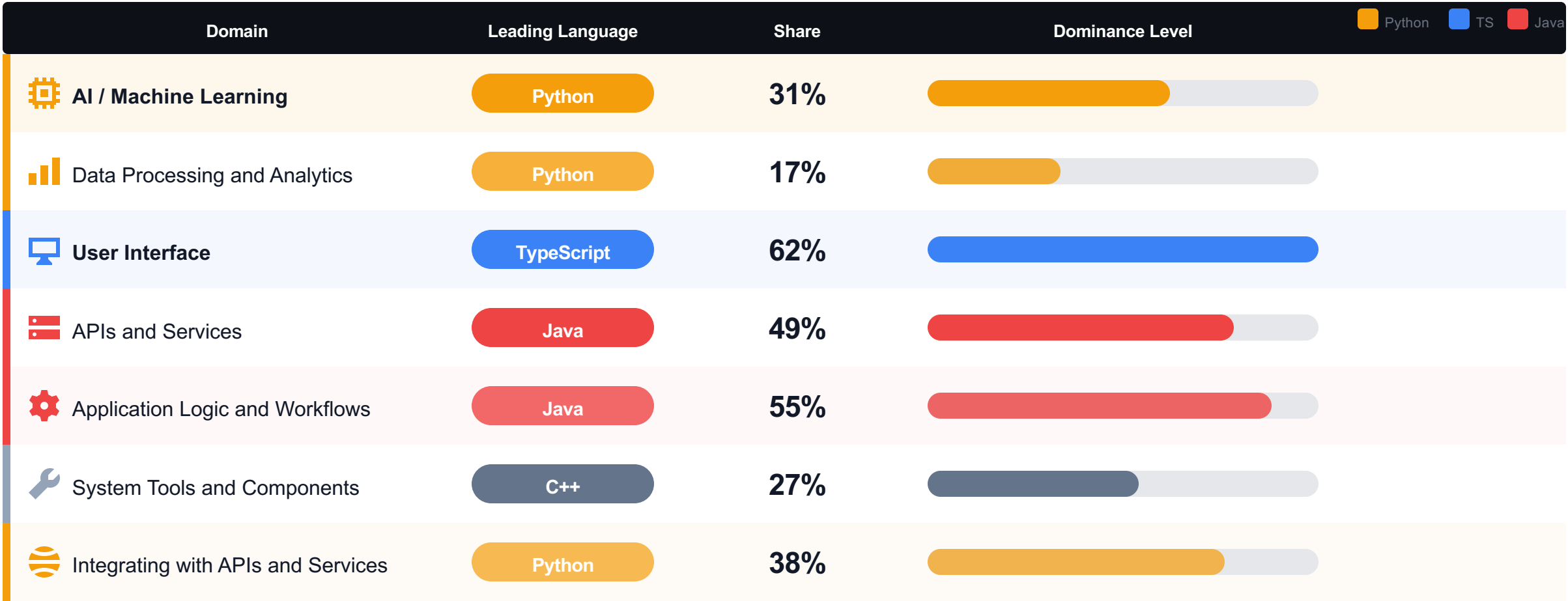
2.3×

TypeScript scores
2.3× higher than C

199 vs 86

Language choice is driven by domain specialization, not general preference — each language owns its niche

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript



Takeaway: Python dominates 3 of 7 domains (AI, data, API integration). TypeScript has the highest single-domain share at 62% (UI).

What This Means for Your Technology Strategy in 2025

01 Invest in Python for AI/ML

Python is the undisputed leader in AI/ML (31% share) and data science (28%). Every AI initiative starts with Python.

31% AI/ML domain share

02 TypeScript Is the Future

TypeScript leads UI (62%), tops satisfaction (199), and tripled usage since 2017. Full-stack TypeScript is the clear trend.

62% UI domain share

03 Rust and Go for Infra

Both are growth plays for cloud-native and systems programming. Go stabilized at 18%; Rust climbs steadily to 11%.

+198 Rust satisfaction score

04 PHP and Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17% (2017-2024); Ruby fell from 10% to 4%. Plan migration paths now. Talent pools are thinning and new developers are choosing TypeScript and Python instead.

-13pp PHP usage decline since 2017

05 Satisfaction = Retention Strategy

Developer satisfaction correlates with type safety and modern tooling. The most loved languages (TypeScript, Rust) offer strong type systems and excellent DX. Factor language choice into your retention strategy.

199 TypeScript satisfaction — highest of any language

DATA-DRIVEN TECHNOLOGY STRATEGY STARTS HERE